DEPARTMENT OF AGRICULTURE IV-A · DEPDev IV-A

Regional Food Insecurity Risk Forecast

CALABARZON · Q4 2025 (Oct-Dec)

Generated 29 September 2026 from aiPHeed model output.

What this number is

The modelled probability that this area falls in the HIGH class of the composite food stress label. **It is an indicative relative risk estimate, not an official food insecurity classification**, and it is published at the Maximum Reliable Lead Time the model established empirically.

Headline

FOOD INSECURITY RISK PROBABILITY

0.499

LOW · quarter-on-quarter +0.0000

Provinces covered	**5 of 5**
Forecast horizon	**Same-quarter estimate. No forward lead time met the 75% reliability threshold of Section 3.4.4, so none is published.**
Provinces above cutoff	**2**
Provinces withheld	**0**
Articles analysed	**3 (0 located)**

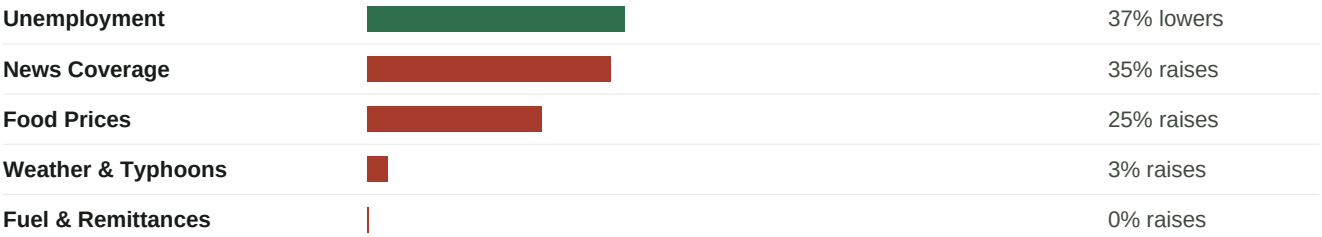
The regional figure is the mean of 5 published province scores. The model has no region-level series of its own. A province withheld by a reviewer is excluded from the mean rather than counted as zero.

Provinces compared

Province	Probability	Level	Class	QoQ	Poverty	Articles
Laguna	0.5279	HIGH	HIGH	+0.0000	4.2%	0
Rizal	0.5279	HIGH	HIGH	+0.0000	3.7%	0
Batangas	0.4847	LOW	LOW	+0.0000	7.1%	0
Cavite	0.4774	LOW	LOW	+0.0000	4.2%	0
Quezon	0.4774	LOW	LOW	+0.0000	16.6%	0

Poverty incidence and availability shock are different signals and do not track together. A province may hold the region's highest poverty rate and its lowest modelled risk in the same quarter.

What is driving the regional figure



Red bars raise the predicted shortfall, **green** lower it. A large bar is not automatically bad news -- share of the explanation and direction of effect are separate things.

For CALABARZON in 2025-Q4, the model's prediction breaks down as: Unemployment 37%, News Coverage 35%, Food Prices 25%, Weather & Typhoons 3%, Fuel & Remittances 0%. Unemployment is the largest single contributor, holding the score down. News Coverage and Food Prices clear the 20% line while raising the score. Unemployment clears the 20% line while lowering it. Shares are SHAP contributions to the province's food insecurity risk probability, recomputed every quarter.

News evidence

3 relevant articles were analysed for this quarter. Of these, 0 could be located to a specific province; the remaining 3 mention the region without naming one, so they inform the regional picture but no individual province.

Strongest-signal headlines

Date	Headline	Score
2025-12-06	**Agri chief seeks gradual, strategic return of rice import tariff to 35%** gmanetwork.com · Unclassified	0.99
2025-10-04	**Philippine eyes temporary onion import ban by February 2026** philstar.com · Unclassified	0.91
2025-11-09	**DA denies 'ghost deliveries' in fertilizer subsidy, calls it fake news** sunstar.com.ph · Unclassified	0.82

Score is the classifier's food-insecurity relevance for the article, 0-1. The corpus is machine-selected and not hand-verified; treat individual headlines as leads rather than findings.

Method

The model predicts, for each CALABARZON province, the probability that the province falls in the HIGH class of the composite food stress label -- SWS quarterly hunger prevalence combined with the province's food CPI deviation from the regional mean, thresholded at the global median. Inputs are the bias-corrected Food Stress Sentiment Index and its lags and acceleration, the five bilingual trigger proportions, and government indicators from PSA, BSP, DOE, PAGASA and PhilRice.

Measure	Value
Maximum Reliable Lead Time	0 quarter(s)
Accuracy	0.6000
Precision (weighted)	0.9200
Recall (weighted)	0.6000

Measure	Value
F1 score (weighted)	0.6762
AUC-ROC	0.9444
F1 target (>= 0.75)	FAIL
AUC-ROC target (>= 0.80)	PASS
Features in the model	12
Baseline: aipheed lgbm combined	0.6727
Baseline: full feature logistic	0.5455
Baseline: lgbm nlp only	0.6182
Baseline: lgbm psa only	0.7273
Baseline: naive persistence	0.8000
Baseline: nlp only logistic	0.5455
Baseline: psa only logistic	0.5818

Accuracy must be read against its baselines. The comparison models above are evaluated on the same walk-forward folds at the same forecast gap, so the figures are directly comparable. Naive persistence in particular is a demanding baseline on an autocorrelated label: a model that does not clearly beat it is not earning its complexity.

Limitations

Indicative risk, not official classification

Outputs are relative ordinal estimates of potential risk. They are not definitive determinations of food insecurity status and must not be read as official government assessments or policy recommendations.

Proxy label, not a direct measurement

No publicly accessible province-quarter food insecurity label exists in the Philippines. The training target is a composite of SWS hunger prevalence, which has no province-level resolution, and PSA food CPI deviation, which reflects supply-side price conditions rather than household hunger experience directly.

Reviewed before publication

Province-quarter figures pass through analyst review. A withheld figure is excluded from this report and from the regional average.

Scope of this assessment

This assessment reports findings only. It does not recommend interventions: no validated mapping exists between this indicator and a response protocol. Pair it with the Department's own standing guidance.